

Machine – Calibrate

Menu: Machine > Calibrate

Use this screen to determine the **Cal Factor** (pulses per unit) for a product. The goal is to achieve a consistent meter roller RPM or liquid flow rate matching typical field operation.

Step-by-Step Calibration

① Step 1 — Set Ground Speed

- Enter the ground speed you typically operate at.

② Step 2 — Set Meter Roller RPM / Valve Position

- Switch the product **ON** using the power button.
- Enter an **Initial Base Rate** and an **Estimated Cal Factor**.
- Press **Start** — RC adjusts flow to match the target rate.
- If successful, RPM is locked.
- The PWM value used is shown in the **PWM** box (for motor controllers). While locked and stopped it can be edited — accept the value RC found, or enter your own (0–255).
- RC stops automatically when the target rate is met.

If RC fails to reach target RPM:

- Press **Stop**.
- Check ground speed.
- Adjust the estimated Cal Factor, PWM min/max, or metering drive range.

③ Step 3 — Set Actual Cal Factor

- Press **Start** to run the meter roller and collect a sample.
- Press **Stop**, then enter the measured amount collected.
- RC calculates and displays the new Cal Factor.
- Press **Save** to record it.

④ Step 4 — Optional Adjustments

- To redo meter speed: press **Unlock**, then repeat Step 2.
- To reuse a previously saved setting: press **Lock**, then proceed to Step 3.
- To skip Step 2 entirely (motor controllers): press **Lock**, enter the desired **PWM** directly, then proceed to Step 3. Useful for iterating on the metering-unit opening at a fixed PWM — run, weigh, adjust the opening, repeat.

Controls

Control	Action
 Start	Start the meter roller / begin collecting a sample
 Stop	Stop the meter roller
 Lock/Unlock	Lock or unlock the current RPM setting
 Power on/off	Switch the product on or off
 Cancel	Discard changes
 Save	Save the calculated Cal Factor
PWM	PWM used while locked (motor controllers only). Editable when locked and stopped; press Save to keep the value
Hz	Live sensor pulse frequency for the selected product (shown for all control types)

Important Notes

- AgOpenGPS is **not** used during calibration.
- The Switchbox is **not** used during calibration.
- If a switchbox is present, ensure the **master switch is OFF** before calibrating.
- Calibration uses **Constant UPM, Mode 2**.
- All sections on the calibrating module are activated during calibration; modules that are not calibrating are unaffected.
- More than one product can be calibrated at the same time, on the same or different modules (requires up-to-date module firmware).
- After calibration, the original mode and section states are restored.
- Start is refused if a locked motor product has PWM 0 — enter a PWM value first, or the motor would not turn.
- Watch the **Hz** display while running: aim for at least 10–12 Hz at working rate for a stable reading. Very high frequencies can exceed the sensor's switching limit — if Hz plateaus while speed rises, the sensor is at its ceiling.